Experiment 4

```
clc;
clear;
close all;

% Simulation of Zigbee, 6LoWPAN and NB-IoT
% Performance Evaluation for IoE Applications

% Protocol names
protocols = {'Zigbee', '6LoWPAN', 'NB-IoT'};

% Simulation parameters
packets = 1000;
packetSize = 1000; % bits

% Assumed performance parameters for demonstration
PDR = [95 92 98];           % Packet delivery ratio (%)
delay = [20 35 80];          % Average delay (ms)
energy = [0.5 0.7 1.2];      % Energy consumption (J)
throughput = (PDR/100) .* packets .* packetSize / 10;

% Display results
fprintf('\nPerformance Comparison of IoE Protocols\n');
fprintf('----------------------------------------\n');

for i = 1:length(protocols)
   fprintf('\nProtocol: %s\n', protocols{i});
   fprintf('Throughput: %.2f kbps\n', throughput(i));
```

```
    fprintf('Average Delay: %.2f ms\n', delay(i));
    fprintf('Packet Delivery Ratio: %.2f %%\n', PDR(i));
    fprintf('Energy Consumption: %.2f J\n', energy(i));
end

% Plot throughput
figure;
bar(throughput);
set(gca, 'XTickLabel', protocols);
ylabel('Throughput (kbps)');
title('Throughput Comparison');
grid on;

% Plot delay
figure;
bar(delay);
set(gca, 'XTickLabel', protocols);
ylabel('Delay (ms)');
title('Average Delay Comparison');
grid on;

% Plot Packet Delivery Ratio
figure;
bar(PDR);
set(gca, 'XTickLabel', protocols);
ylabel('PDR (%)');
title('Packet Delivery Ratio Comparison');
ylim([0 100]);
grid on;

% Plot energy consumption
```

```
figure;
bar(energy);
set(gca, 'XTickLabel', protocols);
ylabel('Energy (J)');
title('Energy Consumption Comparison');
grid on;

% Combined performance comparison
figure;
subplot(2,2,1);
bar(throughput);
set(gca, 'XTickLabel', protocols);
title('Throughput');
ylabel('kbps');
grid on;

subplot(2,2,2);
bar(delay);
set(gca, 'XTickLabel', protocols);
title('Delay');
ylabel('ms');
grid on;

subplot(2,2,3);
bar(PDR);
set(gca, 'XTickLabel', protocols);
title('Packet Delivery Ratio');
ylabel('%');
grid on;

subplot(2,2,4);
```

```
bar(energy);
set(gca, 'XTickLabel', protocols);
title('Energy Consumption');
ylabel('J');
grid on;

sgtitle('Zigbee vs 6LoWPAN vs NB-IoT');
```

Result :

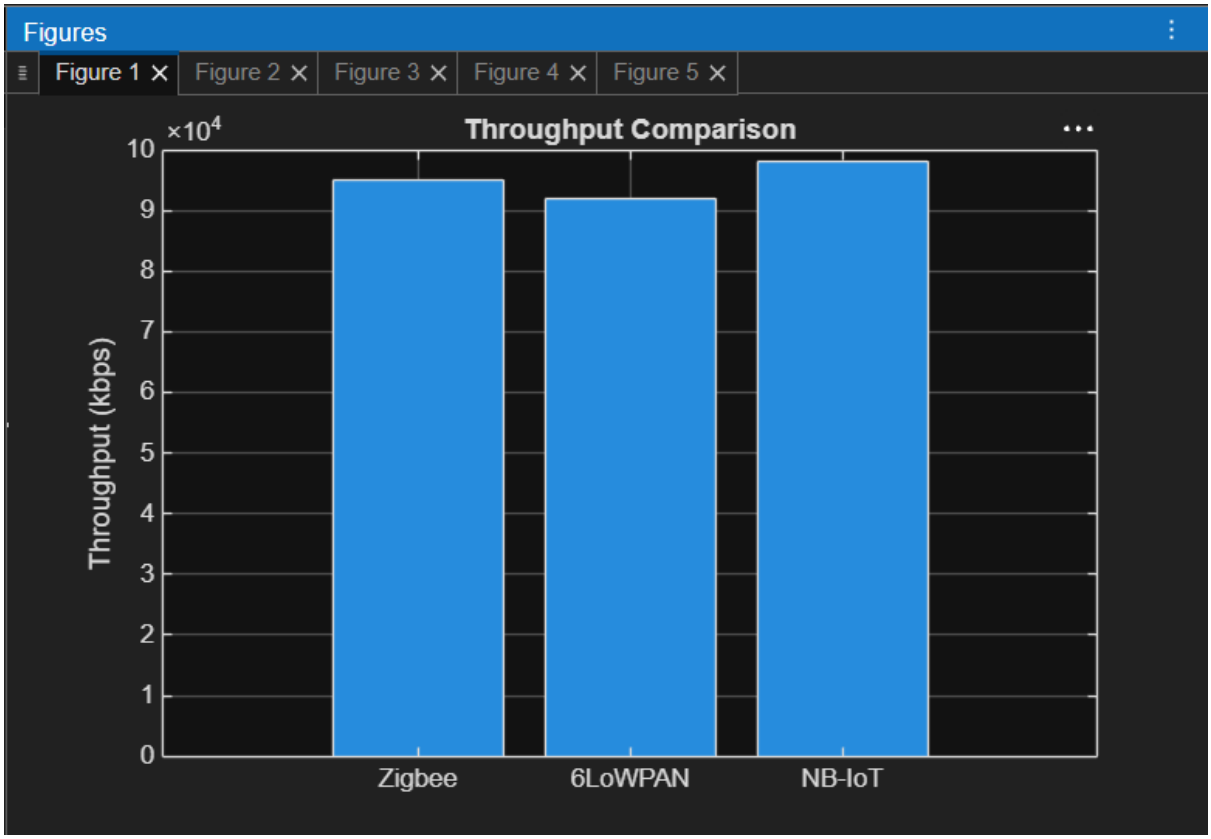
Performance Comparison of IoE Protocols

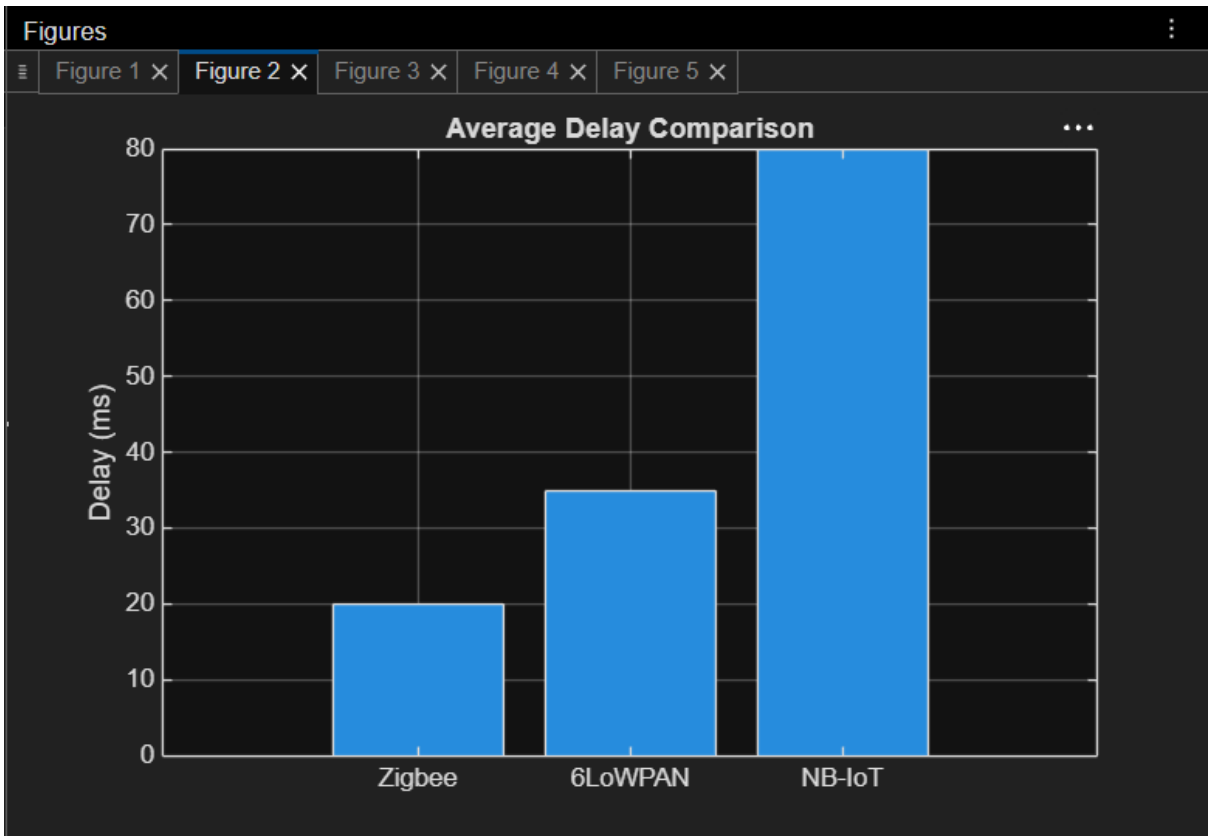
Protocol: Zigbee
Throughput: 95000.00 kbps
Average Delay: 20.00 ms
Packet Delivery Ratio: 95.00 %
Energy Consumption: 0.50 J

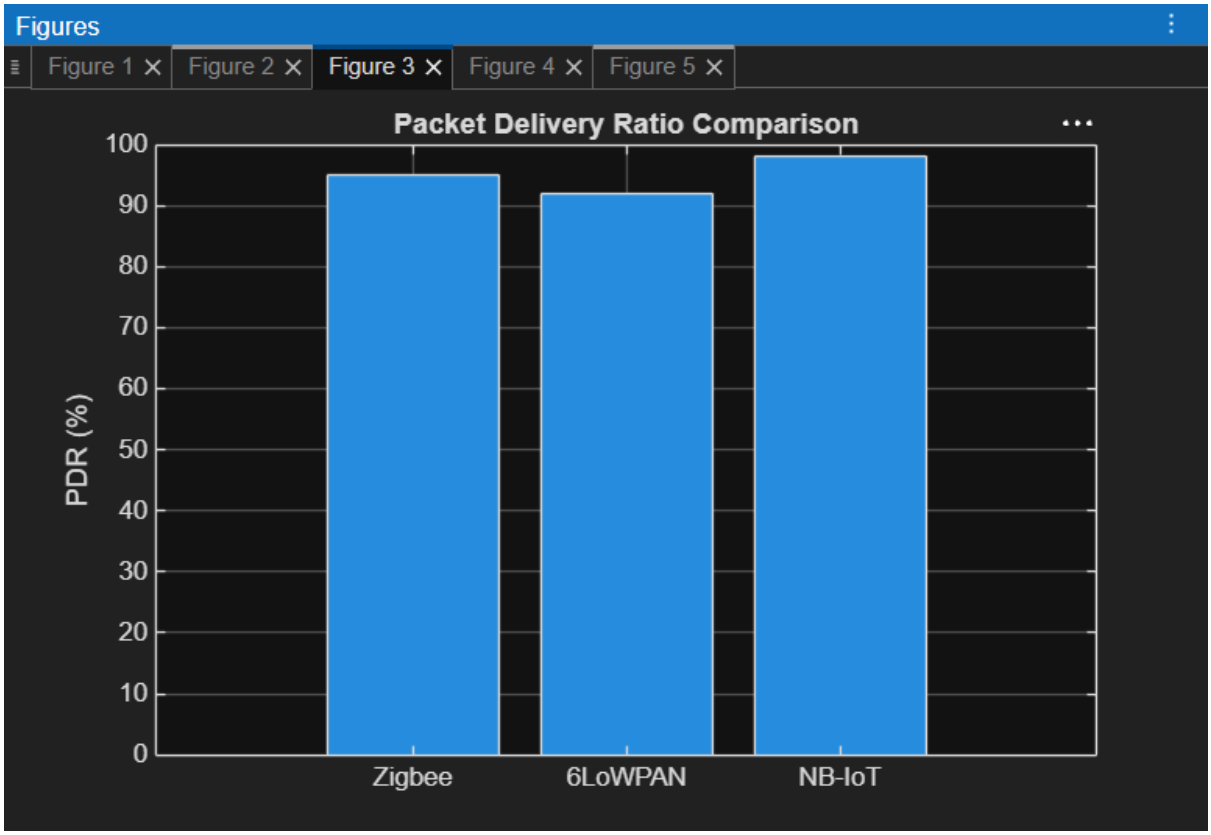
Protocol: 6LoWPAN
Throughput: 92000.00 kbps
Average Delay: 35.00 ms
Packet Delivery Ratio: 92.00 %
Energy Consumption: 0.70 J

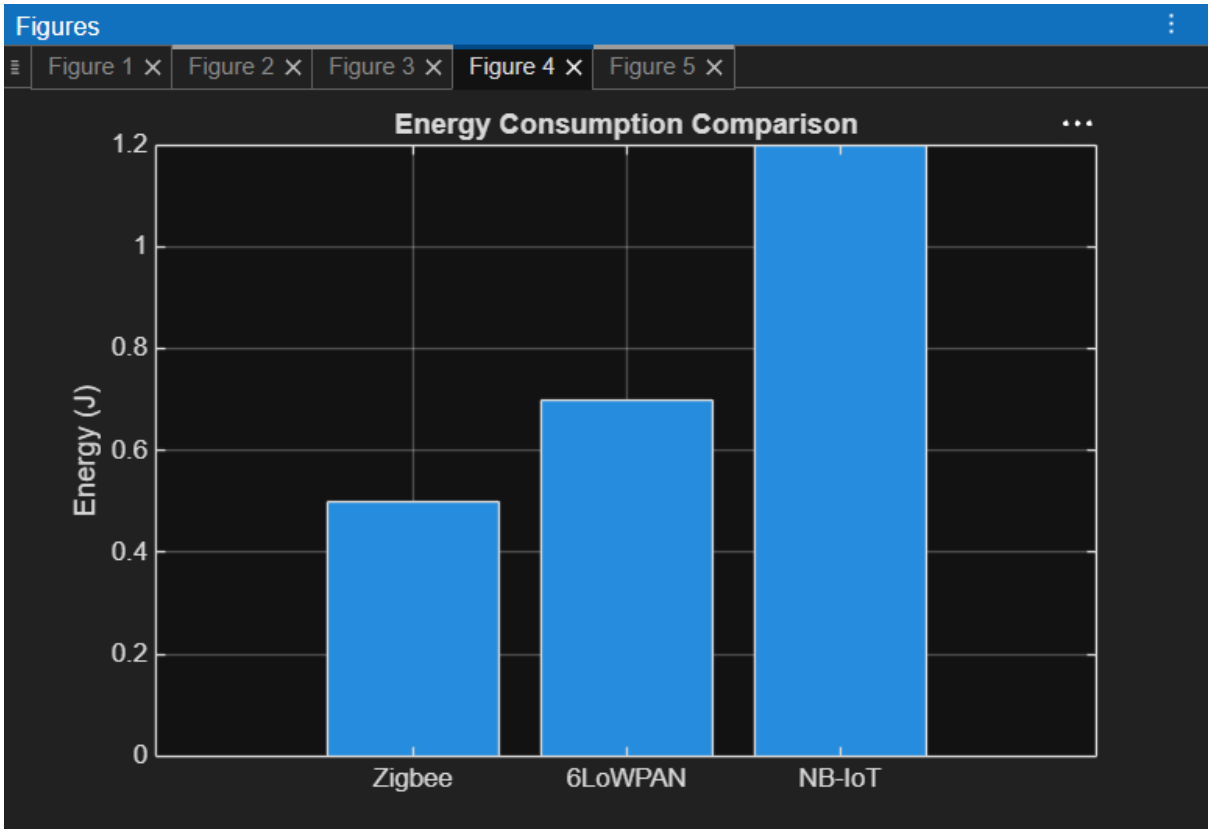
Protocol: NB-IoT
Throughput: 98000.00 kbps
Average Delay: 80.00 ms
Packet Delivery Ratio: 98.00 %
Energy Consumption: 1.20 J

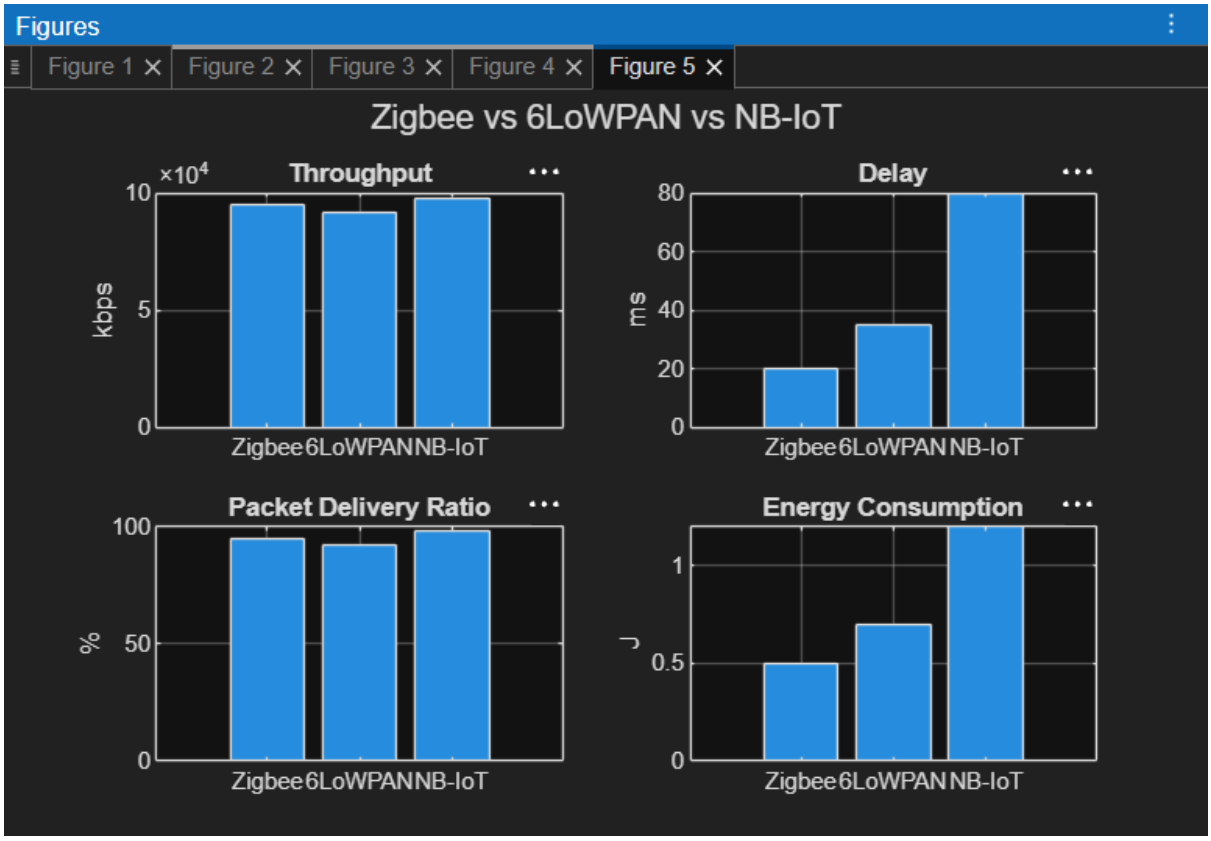
>>


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Throughput Comparison
×10^4
Throughput (kbps)
0
1
2
3
4
5
6
7
8
9
10
Zigbee
6LoWPAN
NB-IoT


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Average Delay Comparison
Delay (ms)
0
10
20
30
40
50
60
70
80
Zigbee
6LoWPAN
NB-IoT


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Packet Delivery Ratio Comparison
PDR (%)
100
90
80
70
60
50
40
30
20
10
0
Zigbee
6LoWPAN
NB-IoT


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Energy Consumption Comparison
Energy (J)
1.2
1
0.8
0.6
0.4
0.2
0
Zigbee
6LoWPAN
NB-IoT


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Zigbee vs 6LoWPAN vs NB-IoT
Throughput
×10^4
10
5
0
kbps
Zigbee
6LoWPAN
NB-IoT
Delay
80
60
40
20
0
ms
Zigbee
6LoWPAN
NB-IoT
Packet Delivery Ratio
100
50
0
%
Zigbee
6LoWPAN
NB-IoT
Energy Consumption
1
0.5
0
J
Zigbee
6LoWPAN
NB-IoT